Problem 5

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Problem 1 *Differentiate the following functions:*

(a) *Given* $f(x) = (x^2 + 4x - 7)e^{-x}$, *show* $f'(x) = (11 - x^2 - 2x)e^{-x}$.

Explanation.

$$\begin{aligned} f'(x) &= \frac{d}{dx}\Big(\frac{x^2 + 4x - 7}{e^x}\Big) \\ &= \frac{(2x + 4)e^x - (x^2 + 4x - 7)e^x}{e^{2x}} \\ &= \frac{e^x(2x + 4 - x^2 - 4x + 7)}{e^{2x}} \\ &= \frac{11 - x^2 - 2x}{e^x} \\ &= (11 - x^2 - 2x)e^{-x} \end{aligned}$$

(b) *Given* $g(x) = \dfrac{x^2 + 4x - 7}{e^{-x}}$, *show* $g'(x) = \dfrac{x^2 + 6x - 3}{e^{-x}}$

Explanation.

$$\begin{aligned} g'(x) &= \frac{d}{dx}\Big(\frac{x^2 + 4x - 7}{e^{-x}}\Big) \\ &= \frac{d}{dx}\Big((x^2 + 4x - 7)e^x\Big) \\ &= \frac{d}{dx}(x^2 + 4x - 7)e^x + (x^2 + 4x - 7)\frac{d}{dx}e^x \\ &= (2x + 4)e^x + (x^2 + 4x - 7)e^x \\ &= e^x(2x + 4 + x^2 + 4x - 7) \\ &= e^x(x^2 + 6x - 3) \\ &= \frac{x^2 + 6x - 3}{e^{-x}} \end{aligned}$$